

BUILD WITH AI: CODE FOR COMMUNITIES · GDG INDIA

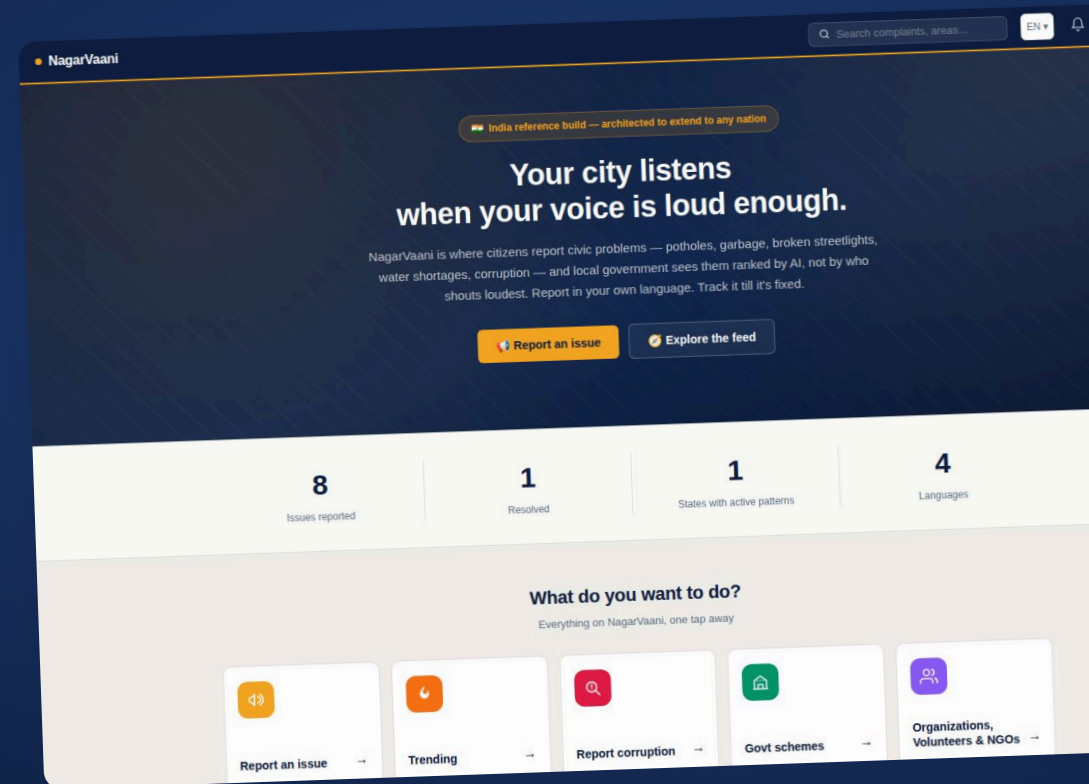
NagarVaani नगरवाणी

Your Voice, Your City.

A civic-intelligence layer that ranks public problems by **how urgent they really are** — not by who shouts loudest.

 <https://nagarvaani-2i2l.onrender.com>

Built by Falguni Takawale · MIT-licensed
github.com/FalguniTakawale/NagarVaani



Citizens speak. Governments can't hear it in one place.



Fragmented

Requests are scattered across helplines, letters, WhatsApp groups, social media and separate portals — nobody sees the whole picture.



The loudest wins

Ranking by votes, recency or city size rewards big cities and busy audiences — and buries villages, minority languages and problems people have stopped hoping will be fixed.



Nothing is measured

Spending drifts away from real need, infrastructure gaps stay open, and there is no way to tell whether an initiative actually changed anything.

The challenge: a scalable, multilingual AI platform that turns citizen requests into priorities policymakers can act on.

OUR INSIGHT

Low votes $\neq$ low importance.

Urgency is a property of the **problem**, not of its audience. A village with one contaminated handpump must outrank a busy road with a hundred upvotes about potholes.

PRIORITY SCORE (0–100) · FROM OUR REAL SCORING CODE, DEMO DATA

 Village handpump, discoloured water · **1 vote**

98.7

 Potholes on a busy road · **104 votes**

40.6

Votes can't outrank danger — they add a small, bounded nudge.

- 1 **Urgency $\neq$ audience.** Whether water is entering homes doesn't depend on who saw the post.
- 2 **Problems repeat.** One blocked drain is often one drain line across five wards — fix it once.
- 3 **Votes serve accountability,** not triage — they show attention and whether issues ever get resolved.
- 4 **Let people speak** — in their own language, by voice or chat.

From a spoken complaint to a ranked action — in five steps.



1 · Intake

Web form, **voice note** or **Telegram**. Hindi, Marathi, Tamil, English — even Hindi typed in English letters.



2 · Filter

AI rejects political or communal framing and suggests a neutral rewording.



3 · Classify

AI reads category, language, location and "is someone likely to be hurt soon?" and translates to English.



4 · Score

Plain arithmetic, no AI.
L1–L5 gives a 0–100 priority you can explain line by line.



5 · Act

Ranked queue, AI brief, map. Status updates reach citizens and voters; a false "resolved" can be disputed.

AI reads. Maths ranks.

The language model never decides who gets help first — that stays transparent and auditable.

Fails open, not closed

If the AI service is down, complaints are still accepted; nothing is lost.

Provider-agnostic

Gemini by default, Claude in one line; Whisper via Hugging Face, Groq or OpenAI.

THE SCORING SYSTEM

L1 – L5: priority you can read.

L1 Safety — could someone be hurt in days? **90–99 band**

L2 Season — monsoon drains, summer water & power **×1.0–2.0**

L3 Type — water 28 · drainage 26 · electricity 22 ... **+10–28**

L4 People — log-scaled population **up to +15**

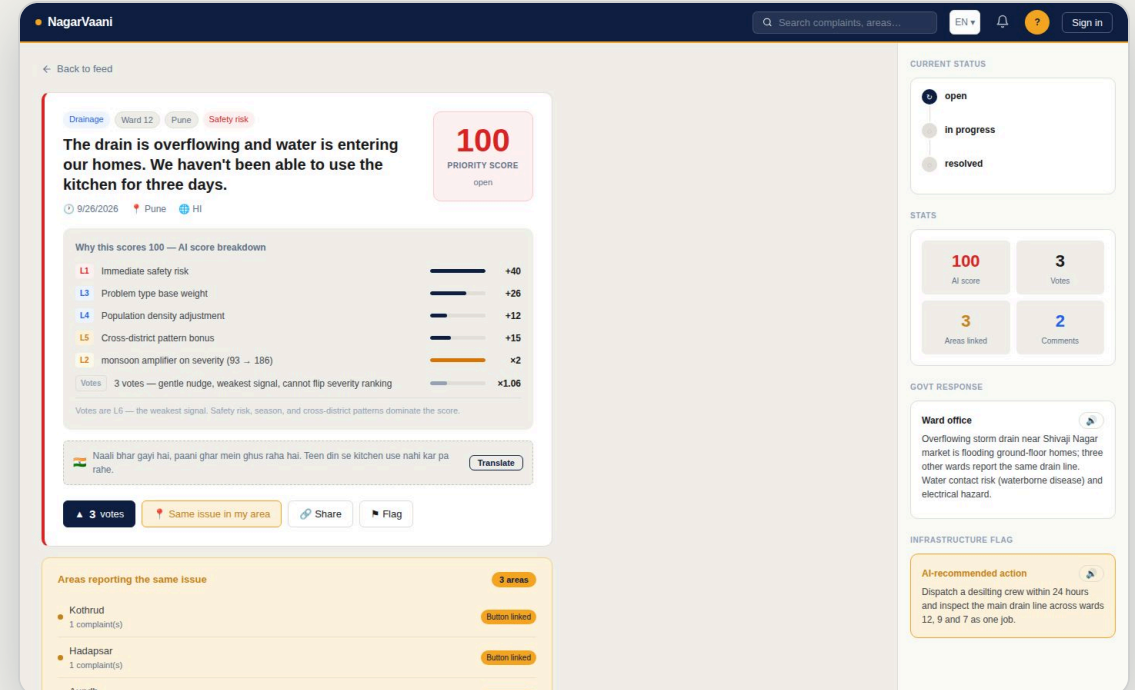
L5 Pattern — same issue in other areas **+5 each, max +20**

Votes — a gentle, bounded nudge **≤ +1 on safety**

score = $\min(100, (L1 + L3 + L4 + L5) \times L2)$

safety risks → 90–99 by severity · then the vote nudge

Season is fixed when first scored — later votes can't change it.



Every complaint shows its own "Why this scores ..." breakdown — and a plain-language *How scoring works* page explains L1–L5 to citizens.

Accountability, not triage.



Similar issues across the whole city

Not only your ward — each with its status — so one locality's problem can be fixed together with the same problem elsewhere in the city.



"See also across the country"

The same problem type in other cities: a pattern visible nationally, not just locally.



Resolution counts

"In Pune: 4 reports of this kind — 1 resolved, 3 still unresolved." Public proof of what gets fixed.



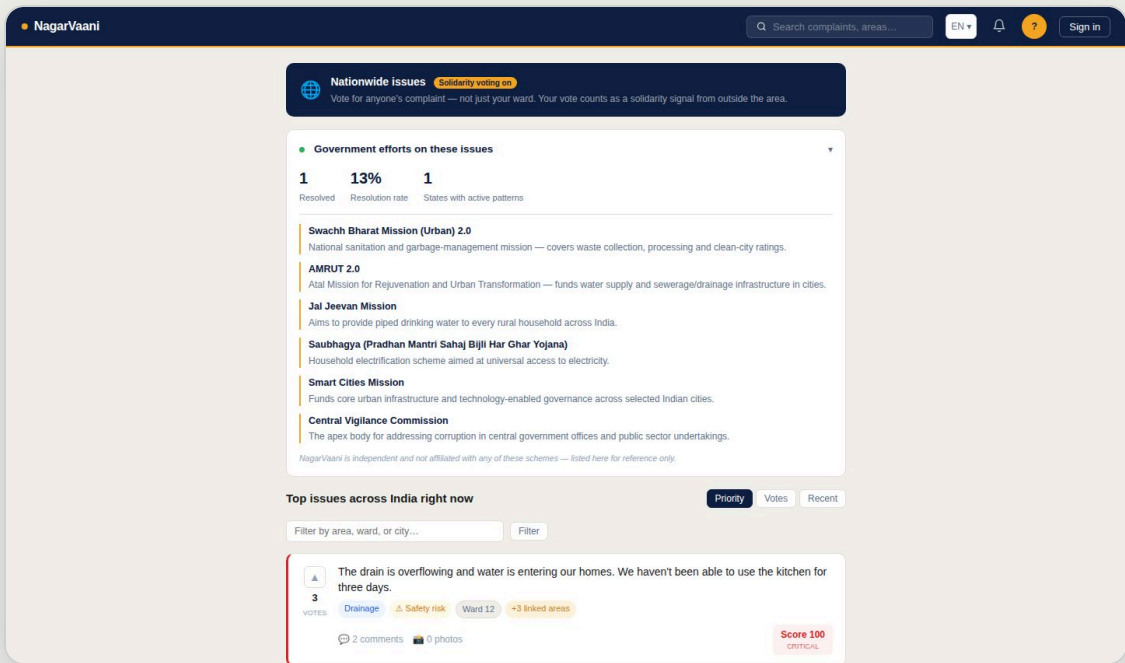
Voters are told when it's "resolved"

...and can say if it isn't. The original reporter can **dispute** a false "resolved".

The screenshot displays the NagarVaani web application interface. At the top, there's a header with the NagarVaani logo, a search bar, and user options like 'EN', a bell icon, and a 'Sign in' button. The main content area shows a complaint thread for a 'Drain overflowing' issue. It includes a status bar at the top indicating 'open', 'in progress', and 'resolved' counts. Below this, there's a 'STATS' section showing '100 AI score', '3 Votes', '3 Areas linked', and '2 Comments'. The 'GOVT RESPONSE' section shows a 'Ward office' response stating that the overflowing storm drain near Shivaji Nagar market is flooding ground-floor homes and that three other wards report the same drain line. The 'INFRASTRUCTURE FLAG' section shows an 'AI-recommended action' to dispatch a desilting crew within 24 hours. The '2 comments' section shows a user comment by Ramesh Kumar and an official reply by Ward Officer Desai. The interface also features a 'Similar issues in your area' section and a 'See also across the country' section. A footer note mentions that areas tagged 'NLP detected' were auto-linked when someone mentioned that place in a comment.

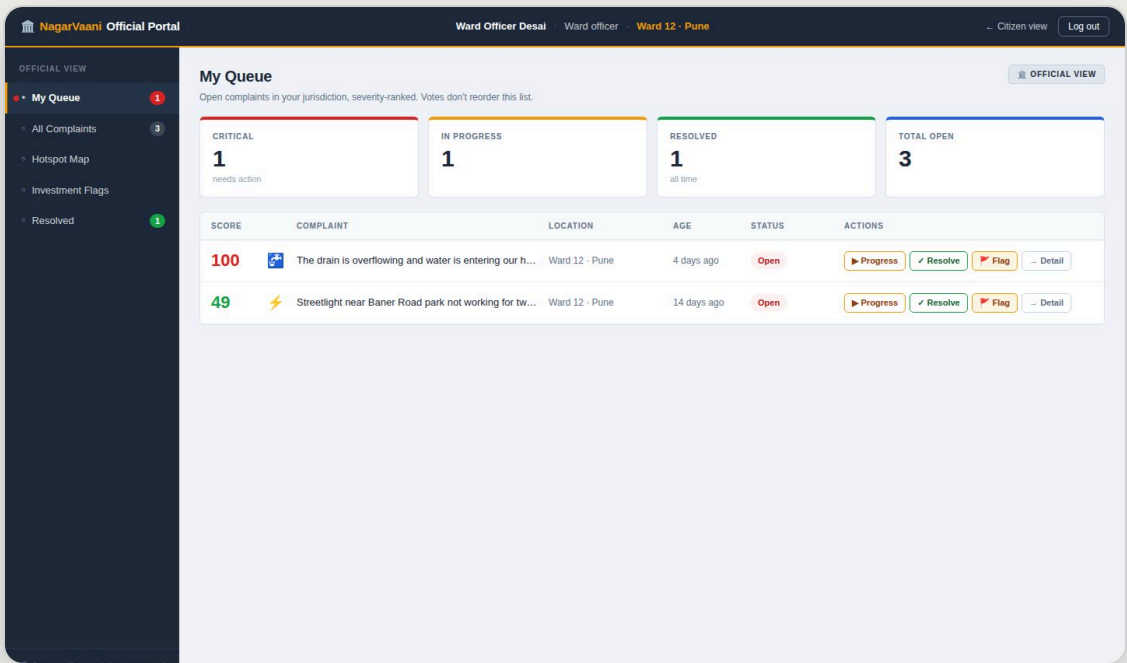
Comment thread with an official reply, automatic place detection ("Kothrud" auto-linked), translation and 🗣️ read-aloud.

Citizens see the nation. Officials see their jurisdiction.



Nationwide issues

Ranked by priority, with solidarity voting from outside the area. Paginated, filterable by area.



Official's queue

Only their ward, city or state — severity-ranked. "Votes don't reorder this list."

Where demand is highest — and which scheme funds it.

NagarVaani Official Portal | Nodal Officer | Central ministry | Nationwide | Citizen view | Log out

OFFICIAL VIEW

- My Queue (4)
- All Complaints (7)
- Hotspot Map
- Investment Flags**
- Resolved (1)
- Official Verifications

Investment Flags
Complaints your office flagged as infrastructure gaps — the list a senior or state-level reviewer works from.

COMPLAINT	SCORE	RECOMMENDED COST	FLAGGED BY	DATE
No flags yet. Use Flag on a complaint in the queue to add it here.				

Data-driven project priorities
Unresolved demand by state and problem type, joined with Census-2011 population and mapped to the real central scheme that funds it. A triage aid — not a costed plan.

[Download CSV](#)

State	Problem	Open	Avg severity	Per million people	Index	Suggested funding scheme
Maharashtra	road	2	68.9	0.018	116.7	PMGSY (rural) / Smart Cities Mission (urban)
Maharashtra	drainage	1	99.6	0.009	99.6	AMRUT 2.0 (urban drainage & water)
Rajasthan	water supply	1	98.7	0.015	98.7	Jal Jeevan Mission / AMRUT 2.0
Maharashtra	tree hazard	1	98.4	0.009	98.4	Municipal horticulture / disaster-management budget
Karnataka	drainage	1	93.3	0.016	93.3	AMRUT 2.0 (urban drainage & water)
Maharashtra	electricity	1	49.4	0.009	49.4	Saubhagya / state DISCOM upgrade

Method: avg severity × (1 + ln(open complaints)); population = Census 2011

Scores reflect severity, season and pattern — not vote count.

Project priorities

Unresolved demand by **state × problem type**, joined with **Census-2011 population**, mapped to the **real central scheme** (AMRUT 2.0, Jal Jeevan Mission, PMGSY...).

Honest by design

A triage index — **not a costed plan**. We removed an early made-up ₹ figure rather than replace it with another.

Take it away

One-click **CSV** download, live hotspot map, investment flags, admin-approved officials.

Not another complaint box.

	TYPICAL PUBLIC-COMPLAINT TOOLS	NAGARVAANI
Ranking	Votes, recency, city size	Severity × season × people × pattern ; votes a small bounded signal
Language	One language, typed	Voice + text, Hindi · Marathi · Tamil · English , Hinglish, two-way translation, read-aloud
The AI	A black-box score	Explainable arithmetic — AI only reads and classifies
Follow-through	Complaint disappears into an inbox	Jurisdiction queue · status log · dispute a false "resolved" · voter notifications
Corruption	Mixed in with potholes	Separate, login-gated channel — never in public feeds
Openness	Closed	MIT licence , swappable AI/speech providers, CSV export

Left column describes common patterns in complaint portals generally, not any single product.

A working platform, not a mock-up.

Citizens

- Report by text, mic, Telegram
- Photos, 360°, GPS → area name
- Anonymous or signed in
- Home · Trending · Near Me · Track
- Vote · comment · "same issue"
- Translate ·  read-aloud · share
- Flag abuse · dispute resolution

45

API operations · 9 routers

Officials

- Admin-approved, work e-mail
- Ward · city · state · central
- Ranked queue & hotspot map
- Status changes inside jurisdiction
- Investment flags
- Project priorities + CSV
- Verification queue (admin)

11

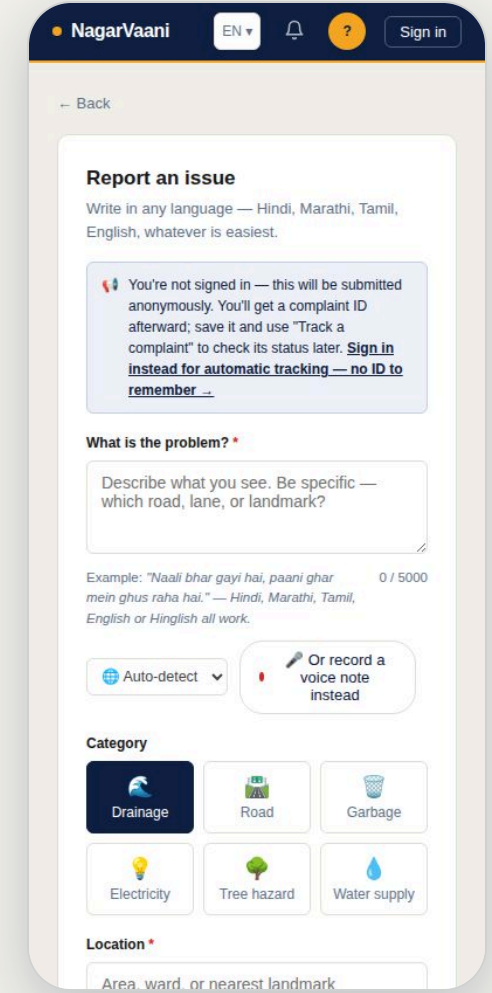
database tables

Platform

- E-mail OTP, reset, Google sign-in
- Rate limits, security headers
- Self-registering Telegram webhook
- E-mail via Brevo + SMTP fallback
- Corruption channel, chatbot
- Neighbourhood e-mail alerts

16

pages, phone-ready

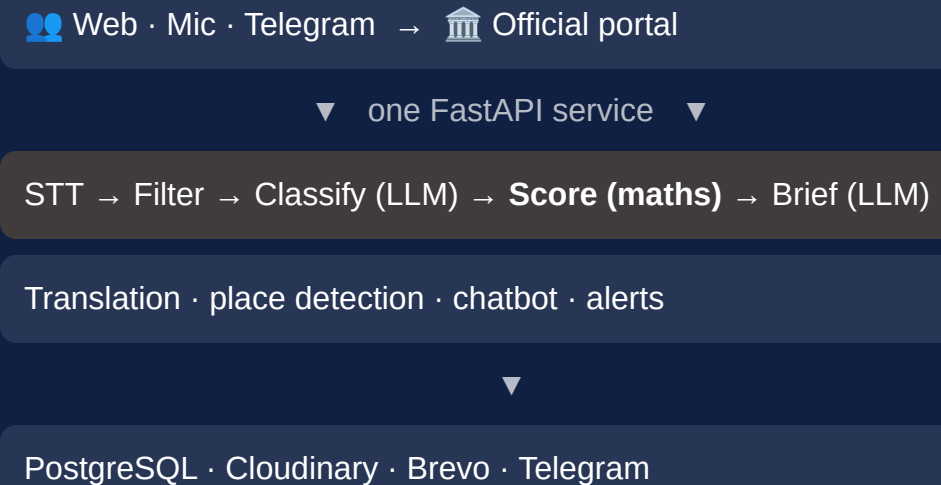


The screenshot shows the NagarVaani mobile app interface. At the top, there's a header with the NagarVaani logo, a language selector (EN), a notification bell, a help icon (?), and a 'Sign in' button. Below the header is a 'Back' button. The main content area is titled 'Report an issue' and includes instructions to write in any language (Hindi, Marathi, Tamil, English). A blue box contains a message about anonymous reporting and a 'Sign in' link. Below this is a text input field for 'What is the problem?' with an example in Hindi. There's a character count '0 / 5000'. Below the text field are two buttons: 'Auto-detect' and 'Or record a voice note instead'. Underneath is a 'Category' section with six icons: Drainage, Road, Garbage, Electricity, Tree hazard, and Water supply. At the bottom is a 'Location' section with a text input field for 'Area, ward, or nearest landmark'.

Simple, swappable, deployable on one free-tier service.

LAYER	CHOICE
Backend	Python · FastAPI · async SQLAlchemy
Database	PostgreSQL (Neon) · SQLite locally
Frontend	Vanilla JS ES modules · Leaflet + OpenStreetMap
Language AI	Gemini (default) · Claude switchable
Speech in	Whisper via Hugging Face / Groq / OpenAI
Speech out	Browser Web Speech API (no key)
Media · chat · mail	Cloudinary · Telegram Bot API · Brevo
Auth	bcrypt · JWT · e-mail OTP · Google (optional)
Hosting	Render (one service) + Neon

Architecture



No framework, no build step — the frontend is served by the same process as the API.

What's real, what's a stub, what's demo.

REAL — proper backend

- Intake: form, voice, Telegram
- AI pipeline & L1–L5 scoring
- Votes, comments, cross-area links
- Status workflow + **jurisdiction enforcement**
- Disputes & e-mail/Telegram notifications
- Auth, OTP, admin-approved officials
- Hotspot map & statistics from the DB
- Priorities + CSV · translation · read-aloud
- Rate limits & security headers

STUB — provision built

- Population factor = default 10,000
(`District` table not wired)
- **No** infrastructure-index / investment data
- **No** project costs — none invented
- Flag moderation: API only, no screen
- Volunteer forms → e-mail only
- "Our Impact" tiles: coming soon
- Schemes page: fixed list

DEMO — labelled

- Seed users & sample complaints
- "Supporters" placeholder cards
- Example cards before data loads

NOT BUILT

- WhatsApp / SMS
- Other countries
- Offline / installable

We attacked our own system — and fixed what we found.

76

scripted endpoint checks — auth, roles, jurisdiction, limits, bad input — all passing

16

pages verified in a real browser at desktop and phone width — zero JS errors

1

critical account-takeover path found and fixed before submission

SEVERITY	FINDING	FIX
Critical	Verification endpoint logged in any verified account; API exposed the author's id	Refuses verified accounts; only an <code>is_author</code> flag is returned
High	No brute-force limits on login / 6-digit OTP	Per-account & per-IP limits (429)
High	An official could change <i>any</i> complaint's status	Enforced inside their jurisdiction
Medium	Forgeable official flag; <code>javascript:</code> image URLs	Reserved prefix; https-only, no inline JS
Medium	Anonymous endpoints could burn AI credit	Per-IP limits, input caps, security headers

Still on the list: token in `localStorage` , in-memory rate limiter, no CSP yet, AI filter fails open.

Strong on intake and prioritisation. Honest about the data gaps.

REQUIREMENT	STATUS	WHERE WE ARE
Voice, text, messaging intake	Done	Web · mic · Telegram (WhatsApp not built)
Multilingual	Done	4 UI languages · AI understands more · two-way translation
Aggregate & surface hotspots	Done	Scoring, cross-area patterns, live map, ranked priorities
Combine with demographic data	Partial	Census-2011 state populations in priorities; per-area density still a constant
Infrastructure indices · investment plans	Gap	Not ingested — the largest gap
Recommend priority projects	Partial	Transparent triage index + matching central scheme; no costs
Measure impact	Partial	Status log, disputes, resolved counts; no outcome dashboard yet
Digital Public Good	Partial	MIT licence · privacy note · swappable providers; open-data export pending
BRICS	Gap	India only — scoring & dashboards are country-agnostic; the data pack is not

Every gap is data or a channel — not a rewrite.

Now — done

- Intake → scoring → officials
- Hotspots, priorities + CSV
- Translation, read-aloud
- Security hardening, deploy

Next — weeks

- Real district population + one infrastructure dataset → **investment-gap** view
- Impact dashboard: time-to-resolve, dispute rate, **neglect index**
- Moderation screen; duplicate merging
- Installable / offline web app

Later — months

- **BRICS:** `country` field + data pack (states, schemes, languages) per nation
- WhatsApp & SMS/IVR channels
- Costed recommendations from schedules of rates
- Open-data export → full Digital Public Good

Hardening on the way: Redis rate limiting · Content-Security-Policy · httpOnly cookies · human-review queue when AI is unavailable.

HOW IT WAS BUILT

One developer. Real decisions. AI as a tool.

NagarVaani was **conceived, specified and driven by one developer** — the problem framing, the scoring philosophy, what to build and what to leave out, deployment and hands-on testing of every flow.

AI coding assistants wrote and reviewed code faster, the way a compiler or linter helps — while the developer set the requirements, rejected what didn't fit, and verified the result.

The judgement shows: we **removed an invented ₹ figure** instead of replacing it, and **fixed a critical flaw** found in our own security review.

TRY IT

<https://nagarvaani-2i2l.onrender.com>

Code & docs: github.com/FalguniTakawale/NagarVaani

Telegram: [@NagarvaaniHackBot](https://t.me/NagarvaaniHackBot)

Licence: **MIT** · Independent civic-tech prototype

Low votes ≠ low importance.